Syllabus Details

Syllabus ID:	10368
Syllabus Name:	**Data Structures and Algorithm_Cấu trúc dữ liệu và giải thuật**
Course Name English:	**Data Structures and Algorithm**
Subject Code:	**CSD201**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h TE + 85' PE + 102.6h self-study
Pre-Requisite:	PRO192
Description:	Upon finishing the course, students can: 1) Knowledge: Understand (ABET e) - the connection between data structures and their algorithms, including an analysis of algorithms' complexity; - data structurre in the context of object-oriented program design; - how data structure are implemented in an OO programming language such as Java 2) Able to (ABET e) - organize and manipulate basic structures: array, linked list, tree, heap, hash - use algorithms for traversing, sorting, searching on studying structures - select a suitable algorithm to solve a practical problem 3) Able to (ABET k) - use JAVA programming language for solving some problems - use Eclipse tool for developing programs in JAVA - Implement some programs in JAVA to solve practical problems based on the studying algorithms 4) Others: (ABET i) - Improve study skills (academic reading, information searching, ...)
StudentTasks:	- Students must attend at least 80% of contact sessions in order to be accepted to the final examination. - Student is responsible to do all exercises given by instructor in class or at home and submit on time - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	358/QĐ-ĐHFPT dated 04/03/2024
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	4/3/2024

4 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Michaelt T. Goodrich, Roberto Tamassia, Michael H. Goldwasser: Data Structures and Algorithms in Java, 6th Edition, 2014	Michaelt T. Goodrich, Roberto Tamassia, Michael H. Goldwasser	Wiley	2014	6th	978-1-118-77133-4	True	False	True	Sách online (Wiley platform)
2	FU slides (ppt)						False	False	False	
3	FU exercises (pdf)						False	False	False	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
4	Code files for students (java files)						False	False	False	

8 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Describe the list data structure and its' different way of implementations. Implement the singly linked list. LO1.1 Demonstrate the list data structure with basic operations: insert, remove, edit, search, sort,... LO1.2 Explain two ways of implementing a list: using arrays and using linked lists. LO1.3 Write programs to implement the singly linked list with basic operations: add (last, first, before, after, at position), search, edit, remove, reverse, sort...) LO1.4 Understand the doubly linked-list data structure with basic operations. LO1.5 Write code snippets which manipulate operations on doubly linked list, and use diagrams to illustrate the effect. LO1.6 Describe the circularly linked-list data structure with some basic operations like add, remove. LO1.7 Expain why the Java code library provides the ArrayList and LinkedList classes
2	CLO2	Define stack and queue. Describe basic operations and the use of these structures. LO2.1 Define stack and queue. LO2.2 List and demonstrate the operations common to stacks and queues. LO2.3 Describe specific tasks to which stacks and queues are suited. LO2.4 Apply stacks to a specific application. LO2.5 Apply queues to a specific application. LO2.6 Write programs to implement stack and queue in Java programming language. LO2.7 Define and explain the need of priority queue.
3	CLO3	Describe about recursive definitions, algorithms, functions and their implementation and use. LO3.1 Describe the basic ideas of recursion and how to set up recursive systems that represent certain real-world phenomena. LO3.2 Know how to develop recursive algorithms and programs LO3.3 Write programs in Java using recursion to solve some problems, like creating the Fibonacci sequence. LO3.4 Analyse a recursive function to find out its' ouput without running. LO3.5 Explain type of recursive functions, give examples and comparing them. LO3.6 Compare recursion with iteration, analyzes their pros and cons
4	CLO4	Explain about general tree, Binary Tree and Binary Search Tree (BST). Implement BST with basic operations. LO4.1 Define general tree, Binary Tree and Binary Search Tree (BST). LO4.2 Given a BST and a sequence of insert and delete operations on it, draw the resulted tree. LO4.3 Find the smallest and largest elements, number of nodes in a tree and its' height. LO4.4 Write code to implement features of a binary search tree, such as insertion, deletion, searching, traversals, nodes and height calculation, rotation ... LO4.5 Derive the time complexities for the above operations on a binary search tree. LO4.6 Compare a binary search tree over other data structures that we have discussed in class. LO4.7 Identify applications where a binary search tree will be useful. LO4.8 Define balanced BST and explane simple balancing algorithm. LO4.9 Define AVL Tree and explain by examples the insertion and deletion operations in it. LO5.10 Define heap and explain its' application.
5	CLO5	Díscuss about graphs and their application. Implement a graph with some basic operations. LO5.1 Demonstrate graphs and their applications. Define an undirected graph and directed graph. LO5.2 Explain basic notions about a graph: vertex, edge, adjacent vertices, incident edge,... LO5.3 Know how to represent a graph using matrix and list LO5.4 Able to apply BFS and DFS to traverse a graph LO5.5 Write prorams to implement Graph with BFS and DFS traversals using Java LO5.6 Explain the shortest path problem and Dijkstra' algorithm. LO5.7 Write program in Java to implement Dijkstra's shortest path algorithm. LO5.8 Define the minimum spanning tree and using the Kruskal's algorithm to find it. LO5.9 Define Euler cycle/path and the necessary and sufficient conditions for their existance. LO5.10 Explain the algorithm to find Euler path / circuit using Stack and implement it in Java lannguage. LO5.11 Define Halmilton cycle/path and able to determine them using backtracking algorithm. LO5.12 Demonstrate the graph coloring and can apply Sequential coloring algorithm to color a graph.
6	CLO6	Explain the operation and performance of some basic and advanced sorting alggorithms LO6.1 Explain the operation and performance of selection sort applying to arrays. LO6.2 Explain the operation and performance of insertion sort applying to arrays. LO6.3 Explain the operation and performance of bubble sort applying to arrays. LO6.4 Explain the operation and performance of quick sort applying to arrays. LO6.5 Explain the operation and performance of merge sort applying to arrays. LO6.6 Explain the operation and performance of heap sort applying to arrays. LO6.7 Demonstrate the operation and performance of radix sort.
7	CLO7	Explain about hashing and application LO7.1 Explain the concept of "hash". Define concepts hash function and hash table and their application. LO7.2 Demonstrate the types of hash functions: Division, Folding,... LO7.2 Explain the collision and collision-handling. LO7.3 Explain the open addressing method for collision-resolution: lear and quadratic probing. LO7.4 Explain the chaining method for collision-resolution: separate chaining and Coalesced chaining. LO7.5 Define perfect hash function and extendible hashing.
8	CLO8	Describe the Text Processing problem and its' application. Explain the Huffman, LZW and Run-length encoding Algorithms. LO8.1 Describe the Text Processing problem and its' application. LO8.2 Explain the Brute Force Text Pattern-Matching algorithm LO8.3 Describe the main idea of The Knuth-Morris-Pratt Algorithm LO8.4 Explain The Huffman Coding Algorithm. LO8.5 Explain steps needed to use The Huffman Coding Algorithm for compressing a text file. LO8.6 Explain the LZW encoding algorithm. LO8.7 Explain the Run-length encoding algorithm.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Introduction 1.1. Using Arrays 1.2. Singly Linked Lists	Offline	CLO1	IT	Text book, slides, exercises and sample practical examples	CSD201	Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
2	1.3. Circularly Linked Lists 1.4. Doubly Linked Lists	Offline	CLO1	IT	Text book, slides, exercises and sample practical examples	CSD201	Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
3	Progress test and/or Review Exercises	Offline	CLO1	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
4	Guiding Exercises/Assignment	Offline	CLO1	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
5	2.1 Stacks	Offline	CLO2	ITU	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
6	Guiding Exercises/Assignment	Offline	CLO2	ITU	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
7	2.2 Queues 2.3 Double-Ended Queues (Deque)	Offline	CLO2	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
8	2.4 The Priority Queue	Offline	CLO2	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
9	Progress test and/or Review Exercises	Offline	CLO2	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
10	Guiding Exercises/Assignment	Offline	CLO2	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
11	3.1 Illustrative Examples 3.1.1 The Factorial Function 3.1.2 Binary Search 3.1.3 File Systems	Offline	CLO3	IT	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
12	3.2 Analyzing Recursive Algorithms 3.3 Further Examples of Recursion 3.3.1 Linear Recursion 3.3.2 Binary Recursion 3.3.3 Multiple Recursion 3.4 Designing Recursive Algorithms 3.5 Eliminating Tail Recursion	Offline	CLO3	IT	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
13	Progress test and/or Review Exercises	Offline	CLO3	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
14	Guiding Exercises/Assignment	Offline	CLO3	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
15	4.1 General Trees 4.1.1 Tree Definitions and Properties 4.1.2 The Tree Abstract Data Type	Offline	CLO4	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
16	4.2 Binary Trees 4.2.1 The Binary Tree Abstract Data Type 4.2.2 Properties of Binary Trees 4.3 Implementing Trees 4.4 Tree Traversal Algorithms	Offline	CLO4	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
17	Progress test and/or Review Exercises Assignment evaluation	Offline	CLO4	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
18	Guiding Exercises/Assignment	Offline	CLO4	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
19	4.5 Binary Search Trees 4.5.1 Searching Within a Binary Search Tree	Offline	CLO4	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
20	4.5.2 Insertions and Deletions	Offline	CLO4	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
21	Progress test and/or Review Exercises	Offline	CLO4	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
22	Guiding Exercises/Assignment	Offline	CLO4	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
23	4.6 Balanced Search Trees 4.7 AVL Trees	Offline	CLO4	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
24	4.8 Heaps	Offline	CLO4	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
25	Progress test and/or Review Exercises	Offline	CLO4	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	
26	Guiding Exercises/Assignment	Offline	CLO4	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 1	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
27	Progress test 1 and review	Offline	CLO4	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
28	Review	Offline	CLO4	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
29	5.1 Graphs 5.2 Data Structures for Graphs 5.2.1 Edge List Structure 5.2.2 Adjacency List Structure 5.2.3 Adjacency Matrix Structure	Offline	CLO5	IT	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
30	5.3 Graph Traversals 5.3.1 Depth-First Search 5.3.3 Breadth-First Search	Offline	CLO5	IT	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
31	Progress test and/or Review Exercises	Offline	CLO5	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
32	Guiding Exercises/Assignment	Offline	CLO5	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
33	5.4 Shortest Paths 5.4.1 Weighted Graphs	Offline	CLO5	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
34	5.4.2 Dijkstra’s Algorithm	Offline	CLO5	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
35	Progress test and/or Review Exercises	Offline	CLO5	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
36	Guiding Exercises/Assignment	Offline	CLO5	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
37	5.5 Minimum Spanning Trees 5.5.1 Prim-Jarn´ık Algorithm	Offline	CLO5	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
38	5.5.2 Kruskal’s Algorithm 5.6. Euler's tour and Euler's cycle	Offline	CLO5	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
39	Progress test and/or Review Exercises	Offline	CLO5	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
40	Guiding Exercises/Assignment	Offline	CLO5	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
41	6.1 Selection-Sort 6.2 Insertion-Sort	Offline	CLO6	IT	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
42	6.3 Bubble-sort 6.4 Quick-Sort	Offline	CLO6	IT	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
43	Progress test and/or Review Exercises	Offline	CLO6	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
44	Guiding Exercises/Assignment	Offline	CLO6	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
45	6.5 Merge-Sort 6.6 Heap-Sort	Offline	CLO6	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
46	6.7 Linear-Time Sorting: Bucket-Sort and Radix-Sort 6.8 Comparing Sorting Algorithms	Offline	CLO6	T	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
47	Progress test and/or Review Exercises	Offline	CLO6	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
48	Guiding Exercises/Assignment	Offline	CLO6	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
49	7.1 Hash Tables 7.2 Hash Functions 7.3 Collision-Handling	Offline	CLO7	IT	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
50	7.4 Load Factors, Rehashing, and Efficiency 7.5 Java Hash Table Implementation	Offline	CLO7	IT	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
51	Progress test and/or Review Exercises	Offline	CLO7	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
52	Guiding Exercises/Assignment	Offline	CLO7	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
53	8.1 Abundance of Digitized Text 8.2 Pattern-Matching Algorithms 8.2.1 Brute Force 8.2.2 The Knuth-Morris-Pratt Algorithm	Offline	CLO8	IT	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
54	8.3 Text Compression 8.3.1 The Huffman Coding Algorithm 8.3.2 The LZW Algorithm 8.3.3 The Run-length Encoding Algorithm	Offline	CLO8	IT	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
55	Progress test and/or Review Exercises	Offline	CLO8	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
56	Progress test and/or Review Exercises	Offline	CLO8	U	Text book, slides, exercises and sample practical examples		Study just learned materials, do writing exercises and redo practical examples and do assignment 2	
57	Assignment evaluation	Offline						
58	Assignment evaluation	Offline						
59	Progress test 2 and review	Offline						
60	Review	Offline						

120 Constructive question(s)

No.	Session No	Name	Details
1	1	CQ1.1	Please list the advantages of Linked List over Array?
2	1	HCM_CQ1.1	How do you insert an element into an array?
3	2	CQ1.2	What are the disadvantages of linked lists?
4	2	HCM_CQ1.2	What are the disadvantages of using an array? List some cases that you think using an array is not optimal.
5	3	CQ1.3	What are difference between Single, Circularly and Doubly Linked List?
6	3	HCM_CQ1.3	How does a linked list solve the problems of an array? (for inserting, deleting an element)
7	4	CQ2.1	What is a stack?
8	4	HCM_CQ2.1	What is a stack? In your daily lives, describe 3 situations in which the stack principle is used.
9	5	CQ2.2	What are the applications of the stack?
10	5	HCM_CQ2.2	What are the applications of the stack in computer science/information technology?
11	6	CQ2.3	What are the 3 primary methods for a stack?
12	6	HCM_CQ2.3	What are the 3 primary methods for a stack?
13	7	CQ3.1	What is a queue?
14	7	HCM_CQ3.1	What is a queue? What would happen if there is no queue in a coffee shop?
15	8	CQ3.2	What are difference between stack and queue?
16	8	HCM_CQ3.2	What are difference between stack and queue?
17	9	CQ3.3	What is a priority queue?
18	9	HCM_CQ3.3	What is a priority queue? (hint: what happens if an ambulance uses a shared gate with other cars/vihicles in a hospital?
19	10	CQ4.1	What is recursive?

No.	Session No	Name	Details
20	10	HCM_CQ4.1	What is recursive? What is/are recursive step(s) in Fibonacci number definition?
21	11	CQ4.2	How many main parts does a recursive algorithm have?
22	11	HCM_CQ4.2	How many main parts does a recursive algorithm have?
23	12	CQ4.3	What are pro and cons of recursion?
24	12	HCM_CQ4.3	What are pro and cons of recursion?
25	13	CQ5.1	What are difference between Binary Tree and Binary Search Tree?
26	13	HCM_CQ5.1	What are differences between a Binary Tree and a Binary Search Tree?
27	14	CQ5.2	Which operation of tree is more efficient than array and linked list?
28	14	HCM_CQ5.2	Which operation of tree is more efficient than array and linked list?
29	15	CQ5.3	List the application of BT to specific problems?
30	15	HCM_CQ5.3	List some applications/softwares/functions/systems using BST as the key element?
31	16	CQ6.1	What is an AVL tree?
32	16	HCM_CQ6.1	What will happen if you insert an ordered array into a BST?
33	17	CQ6.2	What are difference between and Binary Search Tree and AVL Tree?
34	17	HCM_CQ6.2	What are different between and Binary Search Tree and AVL Tree?
35	18	CQ6.3	What is the complexity of a AVL tree search?
36	18	HCM_CQ6.3	What is the complexity of a AVL tree for searching? Is an AVL different from a BST when searching an element?
37	19	CQ7.1	What is heap?
38	19	HCM_CQ7.1	What is a heap? What is the main difference between a min heap and a max heap?
39	20	CQ7.2	What is the similarity and difference between binary tree, AVL tree and heap?
40	20	HCM_CQ7.2	What are the similarities and differences between binary tree, AVL tree and heap?
41	21	CQ7.3	How to calculate balance factor in AVL tree ?
42	21	HCM_CQ7.3	How to calculate balance factor in AVL tree? Show the formula?
43	22	CQ8.1	What is the relationship between the number of edges and the sum of degrees of all vertices in the graph?
44	22	HCM_CQ8.1	What is the relationship between the number of edges and the sum of degrees of all vertices in the graph?
45	23	CQ8.2	What is breadth-first search algorithm?
46	23	HCM_CQ8.2	What is breadth-first search algorithm?
47	24	CQ8.3	What is depth-first search algorithm?
48	24	HCM_CQ8.3	What is depth-first search algorithm? Compare InOrder, PreOrder, PostOrder traversal methods?
49	25	CQ9.1	What is pseudo-graph?
50	25	HCM_CQ9.1	What is a connected-graph?
51	26	CQ9.2	What is a complete graph?
52	26	HCM_CQ9.2	What is a complete graph?
53	27	CQ9.3	What is the formula for calculating the number of edges of a complete graph with n vertices?
54	27	HCM_CQ9.3	What is the formula for calculating the number of edges of a complete graph with n vertices?
55	28	CQ10.1	What is the difference between Dijkstra and Floy algorithms?
56	28	HCM_CQ10.1	What is the difference between Dijkstra and Floy algorithms? (about the input/output and complexities)
57	29	CQ10.2	How to descript the graph on Java?
58	29	HCM_CQ10.2	How to use the graph structure on Java?
59	30	CQ10.3	What are difference between BFS and DFS?
60	30	HCM_CQ10.3	In a real application, such as GG maps, what happens when a user finds the way from A to B?
61	31	CQ11.1	How to color the graph with the least number of colors?

No.	Session No	Name	Details
62	31	HCM_CQ11.1	How to color the graph with the least number of colors?
63	32	CQ11.2	How to find the shortest path between any two vertices?
64	32	HCM_CQ11.2	In a city map, What is the maximum degree of vertices?
65	33	CQ11.3	What is Dijkstra's algorithm?
66	33	HCM_CQ11.3	What is the key idea behind Dijkstra's algorithm?
67	34	CQ12.1	What is the difference between Dijkstra and Floyd algorithms?
68	34	HCM_CQ12.1	Is Floyd algorithm is the combination of n-time Dijkstra's algorithm?
69	35	CQ12.2	How to know if graph has Euler cycle or not?
70	35	HCM_CQ12.2	How to know if a graph has Euler cycle or not?
71	36	CQ12.3	What is a minimum spanning tree?
72	36	HCM_CQ12.3	What is a minimum spanning tree?
73	37	CQ13.1	What are the similarities and differences of Kruskal and Prim Algorithm?
74	37	HCM_CQ13.1	What are the similarities and differences of Kruskal and Prim Algorithm?
75	38	CQ13.2	What is an Euler cycle?
76	38	HCM_CQ13.2	What is different from Euler cycle to Hamilton cycle?
77	39	CQ13.3	What is Hamilton cycle?
78	39	HCM_CQ13.3	Are there any methods/algorithms to determine if a graph has Hamilton?
79	40	CQ14.1	What is the complexity of the select, insert and bubble algorithm?
80	40	HCM_CQ14.1	What is the complexity of the selection, insertion and bubble sort algorithms? How do you find your answer?
81	41	CQ14.2	Which of the 3 algorithms for selection, insertion and bubble is faster? Why?
82	41	HCM_CQ14.2	Which of the 3 algorithms for selection, insertion and bubble is faster? Why?
83	42	CQ14.3	Which sorting algorithms do you know of, which can handle multithreading?
84	42	HCM_CQ14.3	Among the sorting algorithms you know, which can handle multithread programming?
85	43	CQ15.1	What is Quick sort?
86	43	HCM_CQ15.1	What is Quick sort? What is the key idea behind this algorithm?
87	44	CQ15.2	Can you list the names of the sorting algorithms that belong to the divide-and-conquer algorithm?
88	44	HCM_CQ15.2	Can you list the names of the sorting algorithms that belong to the divide-and-conquer algorithm?
89	45	CQ15.3	Which sorting algorithm needs more space to complete the classification?
90	45	HCM_CQ15.3	Which sorting algorithm needs more space to complete the classification?
91	46	CQ16.1	What is Merge sort?
92	46	HCM_CQ16.1	What is Merge sort? How is this algorithm different from Quick sort?
93	47	CQ16.2	Which algorithm for sorting is the fastest? Why? (Selection, Insertion, Bubble, Quick, Merge, Heap, Radix sort)
94	47	HCM_CQ16.2	What are the main weaknesses of radix sort?
95	48	CQ16.3	Which sorting algorithm is best for memory? Why? (Selection, Insertion, Bubble, Quick, Merge, Heap, Radix sort)
96	48	HCM_CQ16.3	Which sorting algorithm is best for memory?
97	49	CQ17.1	What are the advantages and disadvantages of hashing?
98	49	HCM_CQ17.1	What are the advantages and disadvantages of hashing?
99	50	CQ17.2	How to select hash functions?
100	50	HCM_CQ17.2	How to select hash functions? List some principles to select/design a good hash function?
101	51	CQ17.3	What is a perfect hash function?
102	51	HCM_CQ17.3	What is a perfect hash function?
103	52	CQ18.1	How many ways to resolve collisions in hashing?

No.	Session No	Name	Details
104	52	HCM_CQ18.1	How many ways to resolve collisions in hashing? Compare them in term of time and memory complexities?
105	53	CQ18.2	What is the complexity of Brute-Force algorithm?
106	53	HCM_CQ18.2	What is the complexity of Brute-Force algorithm in text pattern matching?
107	54	CQ18.3	What is the complexity of Knuth-Morris-Pratt algorithm?
108	54	HCM_CQ18.3	What is the complexity of Knuth-Morris-Pratt algorithm? What is the role of T[i] array?
109	55	CQ19.1	What Is the difference in lossy vs lossless compression?
110	55	HCM_CQ19.1	What Is the difference between lossy and lossless compression?
111	56	CQ19.2	How to compare the efficiency of different data compression methods?
112	56	HCM_CQ19.2	How to compare the efficiency of different data compression methods?
113	57	CQ19.3	What is Huffman encoding?
114	57	HCM_CQ19.3	What is Huffman encoding? What is the key idea of this method?
115	58	CQ20.1	What is LZW encoding ?
116	58	HCM_CQ20.1	What is LZW encoding ? How does this method differ from Huffman method?
117	59	CQ20.2	What is Run-Length encoding ?
118	59	HCM_CQ20.2	What is Run-Length encoding ? Do you think this algorithm is used to compress the cartoon movies?
119	60	CQ20.3	What is the difference between Huffman and LZW coding techniques?
120	60	HCM_CQ20.3	List some lossy compression methods? Explain why they are used even they cause information loss?

4 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Progress test (PT)	quiz	2	20.0%	>0	Option 1: 20'-40' Option 2 (For Constructivism Approach only): Follow lecturer's proposal		Option 1: essay or multiple choice Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 15-30 if multiple choice Option 2 (For Constructivism Approach only): Follow lecturer's proposal	PT 1 LO1 - LO4 PT 2 LO5 - LO8 Number of questions: 3-8 for each LOx	on paper or using computer	Progress test must be taken right after the last lectures of required material. Instructor has resposibility to review the test for students after graded.
2	Assignment (AS)	on-going	2	20.0%	>0	Option 1: at home Option 2 (For Constructivism Approach only): Follow lecturer's proposal		Option 1: a problem similar to real one Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 1 Option 2 (For Constructivism Approach only): Follow lecturer's proposal	AS1: LO1 - LO3 AS2: LO4-LO6	Individual or team work, guided by instructor, submission by a given deadline	

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
3	Practical Exam	practical exam	1	30.0%	>0	85'		Preferable to be marked by scripts	1-2 small problems	problem(s) to solve by programming (LO1, LO2, LO4, LO6)	supervised by proctor(s) sent by exam board	Automatic or manual evaluating
4	Final exam	Final exam	1	30.0%	4	60'		multiple choice	50	questions distribution: LO1.x: 5 - 8 LO2.x: 5 - 8 LO3.x: 3 - 7 LO4.x: 5 - 10 LO5.x: 5 - 8 LO6.x: 3 - 5 LO7.x: 3 - 5 LO8.x: 3 - 5 more than 70% new questions (for the current semester);	supervised by proctor(s) sent by exam board	